

Placement prep day 4

#1

```
arr = [1,2,2,3,3,4,4,4,4,5,5,5,5,5]
```

```
unique = []
```

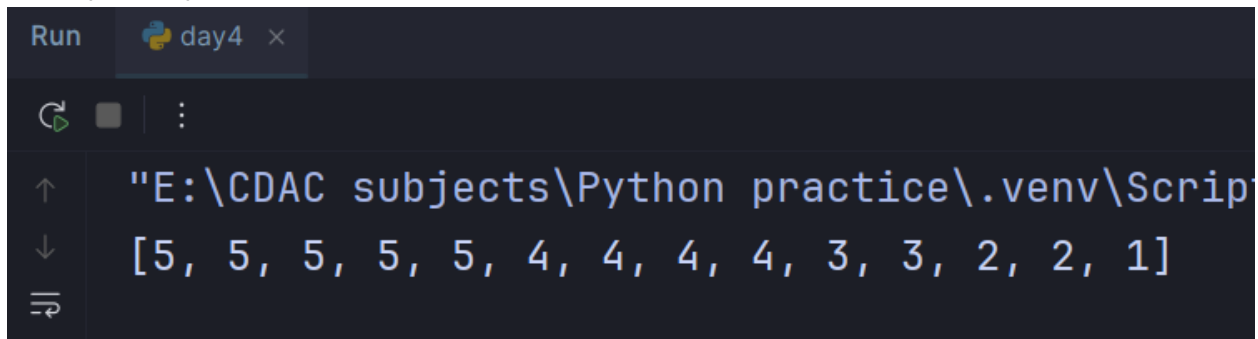
```
for num in set(arr):  
    unique.append(num)
```

```
unique.sort(reverse=True)
```

```
result = []
```

```
for num in unique:  
    result.extend([num] * arr.count(num))
```

```
print(result)
```

A screenshot of a Python IDE terminal window. The window has a title bar with 'Run' and 'day4 x'. The terminal shows the output of the first code block: a list of numbers [5, 5, 5, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 1] printed in reverse order of their frequency in the original array. The terminal also shows the file path 'E:\CDAC subjects\Python practice\.venv\Script' and some navigation icons on the left.

```
Run day4 x  
↑ "E:\CDAC subjects\Python practice\.venv\Script  
↓ [5, 5, 5, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 1]  
⇌
```

#2

```
starts_with = lambda str, ch: str == ch if str else False
```

```
print(starts_with("Hello", "H")) # True
```

```
print(starts_with("Hello", "h")) # False
```

```
Run day4 x
"E:\CDAC subjects\Python
False
False
```

#3

```
words = ["apple", "banana", "cherry", "date", "grapes"]
result = list(filter(lambda x: len(x) == 6, words))
print(result)
```

```
Run day4 x
"E:\CDAC subjects\Python practice\.venv\Scripts
['banana', 'cherry', 'grapes']
```

#4

```
fib = lambda n: [0, 1][:n] if n <= 2 else fib(n-1) + [fib(n-1)[-1] + fib(n-1)[-2]]

print(fib(10))
```

```
Run day4 x
"E:\CDAC subjects\Python practice\.venv\Scr
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```